

has an integrated driver that implements the API, which is used in the module for command and control, and for management data packet traffic. Other details of TCP/IP stack can be found from its data sheet.

Buffer/level shifter. 74HCT125 has quadruple bus buffer gates with 3-stage outputs. These gates feature independent line drivers with 3-state outputs. Each output is disabled when the associated output-enable (OE) input is high. To ensure the high-impedance state during power up or power down, OE should be tied to V_{cc} supply through a pull-up resistor; whose minimum value is determined by current-sinking capability of the driver. The 74HCT125 has been used here as a level-shift interface.

Microcontroller. PIC18F4620 microcontroller, the heart of this project, has features that significantly reduce power consumption during operation.

The key features are alternate run modes and multiple idle modes. The SDO and CS pins of Wi-Fi module are connected to mi-



Fig. 4: MRF24WB0MA module

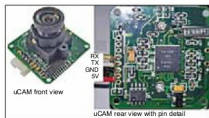


Fig. 5: uCam camera module

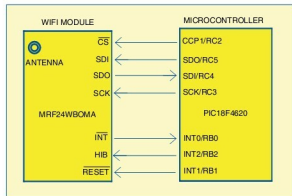


Fig. 6: Interface of MCU PIC18F4620 and MRF24WB0MA

crocontroller through the 74HCT125 buffer.

Microchip's PIC18F4620 was selected because it is low-cost and has sufficient memory for coding—64k Flash memory and 4k RAM. It is available in 40-pin DIP package and has a hardware SPI bus. The code has been written using C18 compiler to fit within the memory of PIC18F4620.

LCD module. 16×2 LCD module has been used to display the IP address and for debugging the project. It is operated in 4-line mode. Port pins RE0, RE1 and RE2 of the microcontroller are connected to RS, R/W and EN control pins of the LCD, respectively. Port pins RD0, RD1, RD2 and RD3 of the microcontroller are connected to D4, D5, D6 and D7 of the LCD, respectively. Preset VR1 is used to control the contrast of the LCD.

MAX232 driver. MAX232 has four sections: dual charge-pump DC-DC voltage converters, RS232 drivers, RS-232 receivers, and receiver and transmitter enable control inputs.

A typical driver output voltage swing is $\pm 8V$ when loaded with a nominal 5-kilo-ohm RS232 receiver and V_{cc} = +5V. Output swing is guaranteed to meet the EIA/TIA-232E and V.28 specifications.

Pins 25 (RC6/TX) and 26 (RC7/RX) of the microcontroller are connected to pins 11 and 12 of MAX232, respectively. Pins 14 and 13 of MAX232 are connected to pins 2 and 3 of DB9 connector, respectively.

Camera. Several camera modules are available in the market. For 5V operation, we chose microCAM (uCAM) serial JPEG camera from 4D Systems (www.4dsystems.com.au) instead of a USB camera to avoid driver-related issues.

The camera is connected to PIC microcontroller through 9-pin DB9 and RS232 level shifter (MAX232). uCAM is a highly integrated serial camera module, which can be attached to any host system that requires a video camera or a JPEG compressed still camera for embedded imaging applications. uCAM comes in a compact form factor with a built-in lens and a 4-wire connector (for 5V, GND, TX and RX) that provides easy access to both power and serial data. It is compatible with C328 JPEG camera. The uCAM used in this project is shown in Fig. 5. You need to connect

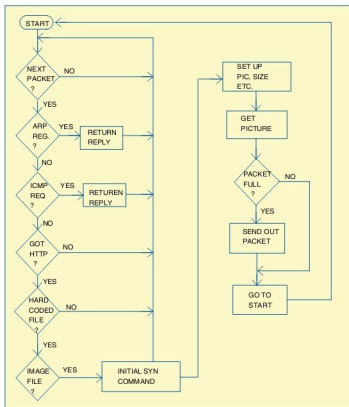


Fig. 7: Flowchart for the program